

Defining Mathematical Functions

1 Expressions and Functions

- functions are expressions
- operations on mathematical functions

2 Step Functions

- discontinuous functions
- deltas and staircases

MCS 320 Lecture 18
Introduction to Symbolic Computation
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symbolic evaluation of expressions

Expressions are callable objects. Evaluation is numeric or symbolic.

- numerical evaluation replaces the input by numbers,
- symbolic evaluation *substitutes* the input by *symbols*.

We can go back and forth between functions and expressions:

- 1 The substitution of an expression
turns an expression into a function.
- 2 The symbolic evaluation of a function
turns a function into an expression.

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derivatives and antiderivatives

We view functions as *mathematical* functions.

- We can compute derivatives with respect to the input variables of a function.

For a function f , which depends on x , the *derivative*

$$\frac{d}{dx}f = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

is the instantaneous rate of change of f at x .

- The *antiderivative* is the opposite of the derivative:

$$\int \frac{d}{dx}f = f + C,$$

for some constant C which does not depend on x .

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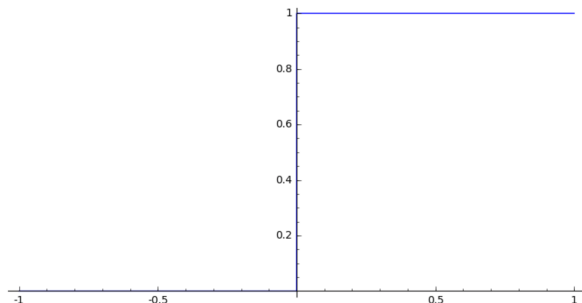
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- **discontinuous functions**
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the unit step function



$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x > 0 \end{cases}$$

Common definitions for $f(0)$ are 0, 1, or $1/2$.

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deltas and staircases

- The derivative of the unit step is the Dirac delta function.

The Kronecker delta is another well known discontinuous function.

- Suppose we want to define a 3-step staircase function $f(x)$, defined by the following four rules:

- 1 $f(x) = 0$, if $x < 0$.
- 2 $f(x) = 1/3$, if $x \in [0, 1)$.
- 3 $f(x) = 2/3$, if $x \in [1, 2)$.
- 4 $f(x) = 1$, if $x \geq 2$.

- The staircase is an example of a *piecewise function*.